MCQ solutions

1. **B.** $P_A(B)$ is the probability of B given A, for $P(A) > 0$. The intersection and the reverse conditioning are different.
2. **B.** $P(A \cap B) = P(A)P_A(B) = 0{,}4 \times 0{,}5 = 0{,}2$. Successive probabilities are multiplied, not added.
3. **C.** By definition of independence, $P(A \cap B) = P(A)P(B)$. An empty intersection corresponds to mutual exclusivity, not to independence in general.
4. **B.** "At most 3" includes 3: $X \leq 3$. "Less than 3" would exclude it.
5. **B.** The probability of a path is the product of successive conditional probabilities. We then add disjoint paths to obtain a union.
6. **A.** The number of trials is 10, the probability of success 0.3, with independence. The variable counts successes, not probabilities.
7. **B.** A sum of n success indicators with expectation p has expectation np. The mean number of failures would be $n(1-p)$.
8. **C.** X is an integer: $X > 4$ is equivalent to $X \geq 5$. $X > 5$ wrongly excludes the value 5.
9. **C.** Independence is one of the assumptions of the Bernoulli scheme. Without it, we can no longer apply the model automatically, even if distributions may happen to coincide in special cases.
10. **A.** Checking the candidate index and the previous one establishes minimality when the probability is monotonic in the size. A single calculation at the candidate index only proves that it works.

True/false. False: with $P(A) = 1/2$, $P(B) = 1/4$ and $B \subset A$, we have $P_A(B) = 1/2$ and $P_B(A) = 1$. True for X integer and k integer: the two events are complementary. False: a single toss of a fair coin has a frequency of 0 or 1, not 1/2.

Solutions to the applications and guidance

Application 1. Row A: 12 and 588; row B: 20 and 380. $P(D) = 32/1000 = 0{,}032 = 0{,}6 \times 0{,}02 + 0{,}4 \times 0{,}05$. $P_A(D) = 0{,}02 \neq 0.032$: no independence.

Application 2. $0{,}14 = 0{,}4 \times 0{,}2 + 0{,}6x$ gives $x = 0{,}1$. $P_D(A) = 0{,}08/0{,}14 = 4/7$.

Application 3. Ill: 90 positive and 10 negative; not ill: 90 positive and 810 negative. $P_M(+) = 0.9$, but $P_+(M) = 90/180 = 0{,}5$.

Application 4. The first is suitable if we count one fixed outcome. Without replacement, the probabilities change and the draws are not independent. The number of trials until the first success is not not fixed: it is not a number of successes in a scheme of fixed length.

Application 5. $1 - P(X \leq 2)$; $P(X \leq 2)$; $P(X \leq 5) - P(X \leq 1)$. $P(X = 0) = (7/10)^8 = 0{,}05764801$.

Application 6. $q(p) = 3p^2(1-p) + p^3 = 3p^2 - 2p^3$. $q' = 6p(1-p) > 0$ on $]0;1[$, therefore q is strictly increasing on $[0;1]$. Continuity and $q(0) = 0 < 0{,}9 < q(1) = 1$ give existence and uniqueness. $q(0{,}80) = 0{,}896 < 0{,}9 < 0{,}905418 = q(0{,}81)$.

Consolidation

Have students keep the table next to the tree; ask at each step "among whom?". For the thresholds, have the integers represented on a number line before calculating. In the parametric problem, distinguish the variable X, the integer $n = 3$ and the parameter p.

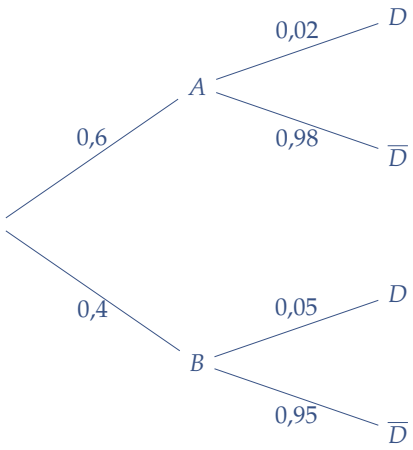
Teaching notes and solutions

Guidance for the teacher

Revisit conditional probabilities in S36 before any Bernoulli scheme. Independence is a model assumption; it does not follow from repetition alone. For drawing without replacement, do not automatically invoke a binomial distribution. The formulas for expectation and variance are used in S39; their proof via a sum of indicator variables is revisited in chapter 14. A threshold required in December must be obtainable with a loop, the logarithm not yet being taught.

Solutions to the activity

1. The tree has the branches A and B, with probabilities 0,6 and 0,4. From A, the branches $D, \overline{D}$ carry 0.02, 0.98; from B, they carry 0.05 and 0.95.



By the law of total probability, $P(D) = P(A)P_A(D) + P(B)P_B(D) = 0{,}012 + 0{,}020 = 0{,}032$.

2. $P(D) > 0$ and $P(A \cap D) = 0{,}6 \times 0{,}02 = 0.012$. Therefore $P_D(A) = \frac{P(A\cap D)}{P(D)} = \frac{0{,}012}{0{,}032} = \frac{3}{8} = 0{,}375$.

3. Each sample is a trial with two outcomes, "defective" or "not defective", with the same probability 0.032 of a defect. The twenty samples being independent, $X \sim \mathcal{B}(20, 0{,}032)$.

4. $P(X = 0) = 0{,}968^{20} \simeq 0.5218$. By the complement,

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1) = 1 - 0{,}968^{20} - 20(0{,}032)0{,}968^{19} \simeq 0{,}1332.$$

Finally $E(X) = 20 \times 0{,}032 = 0{,}64$. This expectation does not imply that a sample has a non-integer number of defects.

5. A common cause linked to the batch could make the defects dependent. The binomial distribution and its calculations rely on the independence given in the model; this assumption should be re-examined for another protocol.

E1. By the law of total probability, $P(B) = 0{,}3 \times 0{,}8 + 0{,}7 \times 0{,}1 = 0{,}31$. Since $P(B) > 0$, $P_B(A) = \frac{P(A\cap B)}{P(B)} = \frac{0{,}24}{0{,}31} = \frac{24}{31}$.

E2. By the binomial formula, $P(X = 3) = \binom{8}{3}(\frac{1}{4})^3(\frac{3}{4})^5 = 56(\frac{1}{4})^3(\frac{3}{4})^5$. The parameters give $E(X) = \frac{8}{4} = 2$ and $V(X) = 8(\frac{1}{4})(\frac{3}{4}) = \frac{3}{2}$.

E3. The complement of $X \geq 4$ is $X \leq 3$, so $P(X \geq 4) = 1 - F(3)$. For the interval, we remove from $X \leq 5$ the values $X \leq 1$, therefore $P(2 \leq X \leq 5) = F(5) - F(1)$.

E4. No, in a standard pack containing four aces. Without replacement, getting an ace on the first draw changes the probability of getting one on the second: the trials are not independent. Moreover, the number of aces can never be 5, whereas a binomial distribution with parameters 5 and $0 < p < 1$ gives a positive probability to this value.

E5. The probability of at least one success is $1 - (\frac{1}{2})^n$. The condition is equivalent to $2^n \geq 100$. Since $2^6 = 64 < 100 \leq 128 = 2^7$, the first integer is 7. The probability is then $\frac{127}{128} \simeq 0{,}99219$.

Solutions to homework DM13

1. X counts the successes in twenty independent Bernoulli trials with parameter 0,8. Hence $X \sim \mathcal{B}(20, 0{,}8)$, $E(X) = 20 \times 0{,}8 = 16$ and $V(X) = 20 \times 0{,}8 \times 0{,}2 = 3{,}2$.

2. The room is short of seats if $X > 18$, i.e. $X = 19$ or $X = 20$. Since these events are mutually exclusive,
$$P(X > 18) = 20(0{,}8)^{19}(0{,}2) + (0{,}8)^{20} \simeq 0{,}069175.$$
The risk is about 6.92 %.

3. The number of refusals R equals 0 for $X \leq 18$, 1 for $X = 19$ and 2 for $X = 20$. By definition of the expectation,
$$E(R) = P(X = 19) + 2P(X = 20) \simeq 0{,}0807045.$$
The takings are $20 \times 10 = 200$ and the payments $30R$, therefore $G = 200 - 30R$. Thus $E(G) = 200 - 30E(R) \simeq 197{,}5789$ monetary units. The 30 already include all the payments related to a refusal; 10 must not be subtracted again.

4. With eighteen bookings, no refusal is possible and the takings are $18 \times 10 = 180$, whatever the number of people present. The expected gain with twenty bookings exceeds this amount by about 17.58 monetary units.

5. The arrivals of people from the same group may be dependent. Moreover, the satisfaction of the audience and the consequences of a refusal are not summed up by the expected gain. The result quantifies a model; on its own it is not enough to dictate a decision.

Solutions to the textbook or past-paper work

The 100 choices are independent, with the same probability $p = 0.0714$. Therefore $X \sim \mathcal{B}(100; 0{,}0714)$.

$$P(X \leq 7) = \sum_{k=0}^{7} \binom{100}{k} (0{,}0714)^k (0{,}9286)^{100-k} \simeq 0{,}577.$$

Finally, $E(X) = 7.14$ and $V(X) = 100 \times 0{,}0714 \times 0{,}9286 = 6{,}630204 \simeq 6{,}63$.